Mohamed Elshamy

Email: elshamy@nmsu.edu
Mobile: +1-575-639-8399
Address: 520 Sweet Avenue, Las Cruces, NM 88001

Visa Status: F1
Personal Website
LinkedIn
GitHub
Google Scholar
ResearchGate

Professional Summary

Ph.D. candidate in Electrical and Computer Engineering at New Mexico State University with 8+ years of industry experience in hardware design, embedded systems, AI, and power electronics. Proven track record delivering production-ready PCBs for industrial, IoT, automotive, and medical applications, with expertise in multi-layer PCB layout, high-speed digital design, power integrity, and EMI-aware design using Altium Designer. Skilled in embedded software development on Nvidia Jetson, PIC, and ESP platforms. Research focuses on power and thermal modeling for SoCs, combining physics-based and machine learning approaches to improve system-level reliability, performance, and intelligent decision-making.

Work Experience

Research Assistant — Las Cruces, NM, USA
New Mexico State University (NMSU) — *Jan 2024 – Present*

- SoC power and thermal modeling, management, and hardware performance optimization
- Physics-informed machine learning and hybrid analytical–ML models
- Machine learning deployment on edge and embedded devices

Senior Hardware Design Engineer — Abu Dhabi, UAE
Tatweer Middle East & Africa - Government Sector — *Jan 2022 – Dec 2023*

- Designing complete PCB solutions, including solar panel MPPT controllers and telematics devices for police cars (OBD diagnostics, tracking, driver behavior monitoring, and power management for speed and seat belt radars).
- Managed the full hardware lifecycle: schematic capture, PCB layout, component selection, fabrication, bring-up, and validation.
- Machine learning–based deployment strategies and performance optimization for radar systems.

Teaching Assistant — Menoufia University, Egypt
Jan 2019 – Jan 2022

- Taught Electronics, Power Electronics, Control Theory, Operational Amplifiers, and more.
- Led labs in PCB Design (Altium Designer) and Embedded C Programming.
- Supported PLC & SCADA systems and industrial automation laboratories.
- Course lectures and labs published on my YouTube channel **[Youtube Link]**

Hardware Design Engineer — Cairo, Egypt
Egyptian Armed Forces Research Center — *Jan 2017 – Dec 2019*

- Served as Manager of the Multi-Layer PCB Laboratory, overseeing fabrication and quality assurance of complex PCBs (up to 8 layers) supporting multiple research and engineering sectors.

Freelance Hardware Design Engineer — Cairo, Egypt
Jan 2014 – Dec 2016

- Designed custom embedded hardware and multilayer PCBs for industrial and IoT applications.
- Managed the full hardware lifecycle: schematic capture, PCB layout, component selection, fabrication, bring-up, and validation.

Education

New Mexico State University (NMSU) — Las Cruces, NM, USA
PhD in Electrical and Computer Engineering (ECE) — *Jan 2024 – Aug 2026 (Expected)*
Major: Electrical and Computer Engineering, CGPA: 4.00/4.00

New Mexico State University (NMSU) — Las Cruces, NM, USA
Master of Science (MSc.) in Electrical and Computer Engineering (ECE) — *Jan 2024 – Spring 2026 (Expected)*
Major: Electrical and Computer Engineering, CGPA: 4.00/4.00

Menoufia University — Shebeen, Menoufia, Egypt
Master of Science (MSc.) in Industrial Electronics and Control Engineering — *Jan 2020 – Dec 2022*
Major: Industrial Electronics and Control Engineering, CGPA: 4.00/4.00

Menoufia University — Shebeen, Menoufia, Egypt
Preparatory Year for M.Sc. Degree in Industrial Electronics and Control Engineering — *Jan 2019 – Jan 2020*
Major: Industrial Electronics and Control Engineering, CGPA: 3.8/4.00

Menoufia University — Shebeen, Menoufia, Egypt
Bachelor of in Power Electronics and Control Engineering — *Jan 2011 – June 2016*
Major: Industrial Electronics and Control Engineering, CGPA: 3.81/4.00

Skills Summary

Programming Languages: Python, C/C++, Embedded C, MATLAB, Verilog HDL
EDA Tools & IDEs: Altium Designer, Kicad, Eagle, LTSpice, Proteus, Labview, Multisim, MATALB, Simulink, Vivado, and STM32CubeIDE
Microcontrollers & FPGA Boards: All Arduino Boards, All ESP32 controllers, ARM MCUs, Xilinx Zybo Z7
Communication Protocols/Interfaces: UART, I2C, SPI, RS-232, RS-485, WI-FI, GSM/GPRS, GPS, USB2.0(FS/HS), Ethernet(10/100 via PHY, RMII/MII, OBD-II, CAN, DDR3L, eMMC, QSPI)
Code Management: Git, Docker, GNU/Linux
Test and Measurements: Signal Generators, Oscilloscopes, Spectrum Analyzer, Digital Multimeters (DMMs), Power Supplies (bench DC, programmable), Logic Analyzers, In-Circuit Debuggers / Programmers, High-speed probing & signal-integrity measurements, Impedance / LCR Meters, Vector Network Analyzer (VNA), and Current Probes (AC/DC).
ML Libraries & Frameworks: Scikit-learn, TensorFlow, Keras, NumPy, Pandas, PyTorch, YOLO, NVIDIA cuML

Selected Publications

- M. R. Elshamy, M. Elahi, A. Patooghy and A. -H. A. Badawy, "CPINN-ABPI: Physics-Informed Neural Networks for Accurate Power Estimation in MPSoCs," **2025** IEEE International Performance, Computing, and Communications Conference (IPCCC), Austin, TX, USA. **[Paper Link]**
- M. Elahi, M. R. Elshamy, A. -H. Badawy and A. Patooghy, "iThermTroj: Exploiting Intermittent Thermal Trojans in Multi-Processor System-on-Chips," **2025** IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS), Lansing/E. Lansing, MI, USA. **[Paper Link]**
- M. Elahi, M. R. Elshamy, A. -H. Badawy, M. Fazeli and A. Patooghy, "Matter: Multi-Stage Adaptive Thermal Trojan for Efficiency & Resilience Degradation," **2025** 26th International Symposium on Quality Electronic Design (ISQED), San Francisco, CA, USA. **[Paper Link]**
- M. Elahi, M. R. Elshamy, A.-H. Badawy and A. Patooghy, "On the Thermal Vulnerability of 3D-Stacked High-Bandwidth Memory Architectures," IEEE Computer Architecture Letters (submitted, major revision). **2025** **[Paper Link]**

- M. R. Elshamy, M. Elahi, A.-H. Badawy and A. Patooghy, “Cluster-BPI: Accurate and Validated Fine-Grained Power Estimation with Application to Thermal Hardware Trojan Detection and Localization in SoCs,” ACM Transactions on Design Automation of Electronic Systems (TODAES) (submitted, Major Revision). **2025** [Paper Link]
- M. Elahi, M. R. Elshamy, A.-H. Badawy and A. Patooghy, “SentinelEdge: An Attention-Based Defense for Real-Time Mitigation of Adversarial Thermal Manipulations in System-on-Chips,” ACM Transactions on Embedded Computing Systems (TECS) (submitted, Minor revision). **2025** [Paper Link]
- M. R. Elshamy, M. Elahi, A. Patooghy and A. -H. A. Badawy, ”Fine-Grained Clustering-Based Power Identification for Multicores,” 2024 IEEE 15th International Green and Sustainable Computing Conference (IGSC), Austin, TX, USA, **2024**. [Paper Link]
- M. R. Elshamy, M. Elahi, A. Patooghy and A. -H. A. Badawy, ”Cluster-BPI: Efficient Fine-Grain Blind Power Identification for Defending against Hardware Thermal Trojans in Multicore SoCs,” **2024** IEEE International Performance, Computing, and Communications Conference (IPCCC), Orlando, FL, USA. [Paper Link]
- M. R. Elshamy, H. M. Emara, M. R. Shoaib and A. -H. A. Badawy, ”p-YOLOv8: Efficient and Accurate Real-Time Detection of Distracted Driving,” 2024 IEEE High Performance Extreme Computing Conference (HPEC), Wakefield, MA, USA, **2024**. [Paper Link]
- M. R. Shoaib, M. R. Elshamy, T. E. Taha, A. S. El-Fishawy and F. E. Abd El-Samie, “Efficient deep learning models for brain tumor detection with segmentation and data augmentation techniques,” Concurrency and Computation: Practice and Experience, vol. 34, e7031, 2022. [Paper Link]
- M. R. Elshamy, B. Abozalam and A. S. Abdelmageed Mahmoud, “Real-time control design and implementation of ball balancer system based on machine learning and machine vision,” Concurrency and Computation: Practice and Experience, vol. 34, no. 27, e7317, Oct. 2022. [Paper Link]
- M. R. Elshamy, E. Nabil, A. S. Abdelmageed and B. Abozalam, ”Stabilization enhancement of the ball on the plate system (BOPS) based on Takagi-Sugeno (T-S) fuzzy modeling,” 2021 International Conference on Electronic Engineering (ICEEM), Menouf, Egypt, 2021. [Paper Link]
- M. R. Shoaib, M. R. Elshamy, T. E. Taha, A. S. El-Fishawy and F. E. Abd El-Samie, “Efficient brain tumor detection based on deep learning models,” Journal of Physics: Conference Series, vol. 2128, p. 012012, Dec. 2021. [Paper Link]
- M. R. Elshamy, E. Nabil, A. Sayed and B. Abozalam, “Enhancement of the ball balancing on the plate using hybrid PD/machine learning techniques,” Journal of Physics: Conference Series, vol. 2128, p. 012028, Dec. 2021. [Paper Link]

Selected Hardware Projects

IOT, POwer electronics, Embedded, and telematics

[Projects Portfolio PDF]

References

Mostafa Abdelrehim, PhD
mabdelre@qti.qualcomm.com
Senior Hardware Engineer
Qualcomm Inc.

Prof. Ahmad Patooghy
apatooghy@ncat.edu
Associate Professor
Department of Computer Systems Technology
North Carolina A&T State University